

kcat via Docker - Quick Reference Guide

What is kcat?

kcat (formerly kafkacat) is a lightweight, command-line tool for reading and inspecting Apache Kafka messages. It is fast, uses minimal resources (~15-20 MB), and is ideal for debugging and monitoring Kafka topics in real time.

Why Docker?

kcat has no native Windows installer (not available on Chocolatey or Scoop). Running it via Docker is the simplest approach — zero installation, just one command. The Docker image is only ~20 MB.

Setup

Step 1: Create PowerShell Profile Alias

Open PowerShell and run:

```
powershell  
notepad $PROFILE
```

Paste the following functions and save:

```
powershell
```

```
# UAT Environment
```

```
function kcat-cloud {  
    docker run --rm -it edenhill/kcat:1.7.1 `   
        -b pkc-rgm37.us-west-2.aws.confluent.cloud:9092 `   
        -X security.protocol=SASL_SSL `   
        -X sasl.mechanisms=PLAIN `   
        -X sasl.username=YOUR_UAT_USERNAME `   
        -X sasl.password="YOUR_UAT_PASSWORD" `   
        @args  
}
```

```
# PROD Environment
```

```
function kcat-prod {  
    docker run --rm -it edenhill/kcat:1.7.1 `   
        -b pkc-rgm37.us-west-2.aws.confluent.cloud:9092 `   
        -X security.protocol=SASL_SSL `   
        -X sasl.mechanisms=PLAIN `   
        -X sasl.username=YOUR_PROD_USERNAME `   
        -X sasl.password="YOUR_PROD_PASSWORD" `   
        @args  
}
```

Step 2: Reload Profile

```
powershell
```

```
. $PROFILE
```

Now `kcat-cloud` and `kcat-prod` are available as shortcut commands.

Epoch Timestamp Reference

kcat uses epoch milliseconds for time-based filtering. Use epochconverter.com to convert dates.

Date (UTC)	Start Epoch (00:00)	End Epoch (23:59)
Feb 23, 2026	1771804800000	1771891199000
Feb 24, 2026	1771891200000	1771977599000
Feb 25, 2026	1771977600000	1772063999000
Mar 2, 2026	1772409600000	1772495999000

5 Essential Commands

1. List All Topics and Brokers

```
powershell
kcat-prod -L
```

Shows all brokers, topics, partitions, and replicas in the Kafka cluster.

2. Read Last N Messages from a Topic

```
powershell
kcat-prod -t com.orbcomm.fe4.prod -C -o -10 -e
```

- `-C` = Consumer mode
- `-o -10` = Last 10 messages
- `-e` = Exit when end of topic is reached

3. Filter Messages by Time Range + Keyword

```
powershell
kcat-prod -t com.orbcomm.fe4.prod -C -o s@1771977600000 -o e@1772063999000 -e | Select-String "JJQA324410002" | Se
```

- `-o s@` = Start time (epoch ms)
- `-o e@` = End time (epoch ms)
- `Select-String` = Filter by keyword (use this instead of `findstr` — no line length limit)
- `Select-Object -First 10` = Show only first 10 matches

4. Export Filtered Results to Notepad

```
powershell

kcat-prod -t com.orbcomm.fe4.prod.masterdata -C -o s@1772409600000 -o e@1772495999000 -e | Select-String "28710138"
```

Saves filtered output to `result.txt` and opens it in Notepad automatically.

5. Export Full Day Data per Partition with Readable UTC Timestamp

```
powershell

kcat-prod -t com.orbcomm.fe4.prod -p 0 -C -o s@1771977600000 -o e@1772063999000 -J | ForEach-Object { if ($_ -match
```

- `-J` = JSON output (includes topic, partition, offset, timestamp)
- `ForEach-Object` = Converts epoch timestamp to readable UTC format
- `-p 0` = Specific partition (change 0-7 for each partition)

Important Tips

Tip	Details
Always use <code>-e</code> flag	Makes kcat exit when it reaches end of topic, instead of waiting forever
Use <code>Select-String</code> not <code>findstr</code>	<code>findstr</code> has an 8000 character line limit, Kafka JSON messages often exceed this
Search by partition	Faster than searching all partitions at once for large topics
Count messages first	Check offset range before dumping large time ranges to estimate size and time

Count Messages in a Partition for a Date Range

```
powershell

kcat-prod -t com.orbcomm.fe4.prod -p 0 -C -o s@1771891200000 -c 1 -f "Start Offset: %o\n"
kcat-prod -t com.orbcomm.fe4.prod -p 0 -C -o s@1771977599000 -c 1 -f "End Offset: %o\n"
```

Difference between End and Start offset = total messages for that partition in that time range.

Common Flags Reference

Flag	Description
<code>(-L)</code>	List metadata (brokers, topics, partitions)
<code>(-C)</code>	Consumer mode (read messages)
<code>(-P)</code>	Producer mode (write messages)
<code>(-t)</code>	Topic name
<code>(-p)</code>	Specific partition number
<code>(-o -N)</code>	Read last N messages
<code>(-o end)</code>	Start reading from end (live tail)
<code>(-o s@EPOCH)</code>	Start reading from timestamp
<code>(-o e@EPOCH)</code>	Stop reading at timestamp
<code>(-e)</code>	Exit when end of topic is reached
<code>(-c N)</code>	Stop after N messages
<code>(-J)</code>	JSON output with metadata
<code>(-f)</code>	Custom format string